



Goal:

The goal of this project is to provide a framework in which you can apply your knowledge of microcontrollers to provide an entertaining experience for users and observers.

Purpose:

The underlying purpose of this project is to give you some experience building an electromechanical widget. We expect that this will involve working with sensors, driving actuators, designing event-driven software, and implementing that software in C on a PIC32 microcontroller. We expect to see all of these elements in every solution. Your lab kit contains sensors and several kinds of display, although you are not limited to using only these. You are limited to an expenditure of \$200.00 / team *of your own money* for all materials and parts used in the construction of your project. Materials from the lab kit and the Cabinet of Freedom do not count against the limit; all other items count at their Market Value.

Background:

A time machine prototype in the SPDL accidentally transported the whole lab back to the late 1800s, a time of steam power not electrical power. You find yourself in the middle of a bustling traveling Karlival filled with steam-powered contraptions, suspicious townsfolk, and ticking gears. No one has seen technology like yours before and they don't trust it. To try and fit in, you must make your own contraptions that look like they belong in a steam powered Karlival.

The Task:

Your task is to design and build a **Steampunk-Themed Karlival Game**, a "Temporal Entertainment Module," (TEM) that bridges the gap between past and future. Each TEM should be visually captivating and mechanically interactive, blending steampunk aesthetics with modern mechatronic systems. The experience should demonstrate to 19th-century Karlival-goers the power of electricity, code, and curious 218ers.

Specifications

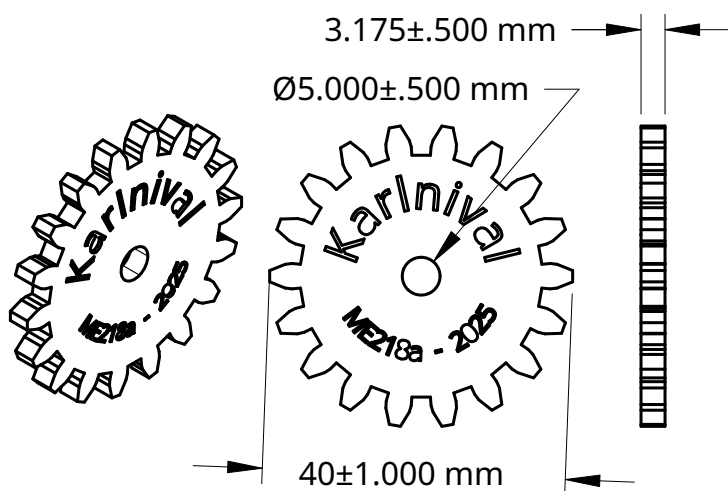
Operation:

- ☐ The TEMs will power up into a welcoming mode. The welcoming mode should encourage the user to interact with the TEM in order to start the game.
- ☐ It should take a player approximately 60 seconds to complete the game with your TEM. Even if a player becomes very skilled at such interaction, it should never take less than 30 seconds.
- ☐ To provide the player with a sense of progress toward task completion, each TEM should include a creative display of the passage of the time elapsed since starting the game.
- ☐ To avoid leaving a game unfinished, each TEM should display an error/timeout condition after 20 seconds of no input from the user. After this, it should reset to the welcoming mode.
- ☐ The game should involve at least 3 distinct inputs from the user and 3 distinct modes of feedback from the TEM.
- ☐ The TEM should be usable without the guidance of an ME 218 engineer or written instructions. Any static instructions must be only in pictorial form (think IKEA instructions).
- ☐ When the TEM determines that the game is complete, it should provide a clear audio, visual, or mechanical indication of such completion. This indication may last no more than 30 seconds before the TEM resets to the welcoming mode. If the game has varying degrees of completeness (e.g. number of points scored), the TEM should display the degree of completeness at this time.

- The TEM must dispense a minimum of 1 and a maximum of 3 **gears** when the game is completed as prizes for the user. Teams can choose how their TEM decides how many gears to dispense. Dispensing gears counts as a mode of feedback and an indication of completeness from the TEM. Dispensing gears is not by itself enough of an indication of completion of the game.

Gears:

- No one at the Karlnival has seen lasercut gears before so they are valuable prizes for TEMs to dispense.
- Gears are made from 3mm (1/8 inch) plywood and have a diameter of 40mm (1.57 inches).
- Gears were designed in Inkscape with Extensions>Render>Gear and the following parameters: teeth=18, tooth size=6.3, pressure angle=18.0, center hole=5mm
- The measurements in the diagram below are approximate. Teams are encouraged to take their own measurements of the gears.



Basic Specifications:

- A team of three or four 218ers will construct a TEM.
- Each TEM must be able to hold at least 50 gears for dispensing to the users.
- The TEM may not be controlled by any microcontroller or processor other than the PIC32MX170F256B.
- While it is permissible to use consumer devices as components in a TEM, in order to avoid intellectual property (IP) issues, such devices must be substantially modified before being incorporated into your project. We don't want you to just buy significant portions of your project. If there is any question as to whether or not the purchased component has been modified significantly enough, please see the teaching staff.
- Each TEM must respond to at least 3 **distinct inputs/interactions** from the user and provide 3 **distinct modes of feedback** to the user
- At least one of the user interactions must be interpreted as an **analog input** to the PIC32 from the user (such as IR, Joy Stick, Potentiometer, etc.)
- The analog input must be used to produce some behavior by the TEM that makes use of the analog nature of the input. No simple thresholds.
- Multiple modes of feedback, including modes not listed here, are encouraged.
- The TEM must have parts that visibly move under the control of the PIC32.

- ☐ At least one of the modes of feedback must be **electromechanical actuation** (such as servo, motor, fan, vibration etc.) and this is in addition to dispensing gears.
- ☐ At least one of the user interactions must involve **non-contact sensing** (such as IR or Audio, etc).
- ☐ Interactions that involve cumulative action (e.g. counting repetitions over a set time period) are encouraged.
- ☐ The complete TEM must be a self contained entity, capable of meeting all specifications while connected only to the provided project power supplies.
- ☐ All components of the TEM **MUST** fit into a total volume no more than 50 cm wide by 50 cm deep by 100 cm high. During operation, the TEM should not require user input from more than 75 cm away from any part of the TEM. Two teams' TEMs must both be usable while sitting together on one of the 1.5 m wide tables in our classroom (550-200).
- ☐ The entire TEM must be easily and safely moved from the SPDL to the grading session and then to the Karl'nival grounds (i.e., to the 550 Atrium) for evaluation. Make sure that you plan for this.
- ☐ While it is normally not a good practice, the finished circuitry may be constructed on your solderless breadboards. This has been done to allow you the maximum time to spend on your project, without having to learn electronic prototyping techniques as well. Be sure to secure the breadboard and wires so that they will not be disturbed when the Karl'nival packs up and travels to the next site.
- ☐ The emphasis in the project is on robust electronics, software, and mechanical systems built with real craftsmanship. Paint alone does not add to either functionality or craftsmanship. This is not to say that you may not decorate the TEM, but simply that it should not become a focus. Any painting that is done near the SPDL must be done using appropriate masking so that no paint residue is left on the building, furniture, sidewalk, driveways, grass, or trees. No painting in the SPDL! And no glitter!
- ☐ Accurate schematics and state diagrams are such a useful aid in debugging that you should be prepared to show your up-to-date schematic or state diagram to a CA whenever you ask them for help on your project.

Safety & Hygiene:

- ☐ The TEMs must be safe for both users and spectators.
- ☐ No glitter!
- ☐ Be considerate of your neighbors in the lab when debugging any audio output; use headphones.
- ☐ There is a strict ban on toxic materials. This prohibition includes Volatile Organic Compounds (VOCs) (i.e. hydrocarbon based spray paints or other noxious fumes). This prohibition also includes while you are working on the TEM in the SPDL.
- ☐ No painting in the SPDL (or anything attached to the SPDL).
- ☐ No part of the TEMs may become ballistic unless completely constrained within the TEM.
- ☐ No pyrotechnics or fire of any kind! Be advised that smoke contains chemicals known to the State of California to cause cancer, birth defects, or other reproductive harm.
- ☐ If the TEMs contain any liquids, they may not be conductive (with the exception of water) or corrosive, and **MUST** be packaged in a fail-safe manner while always being shaken, not stirred.
- ☐ The use of gremlins in your project is ferby-d-den.

Checkpoints

Design Review:

During the day on **11/4/25** in room TBD we will conduct design reviews. Signups for the hour-long slots for 3 teams will happen via a Google Sheet. Each group should prepare a few **simple** PowerPoint slides (scans

of sketches are OK). **No code, no state diagrams, no circuits.** The slides should show your concepts, a preliminary event list with responses, and a list of how you are going to meet the user interface requirements. One member of the team must bring a laptop to connect to the screen for your presentation. You will present these to other members of the class and members of the teaching staff so that all may hear about your ideas and provide feedback and advice. At this time, you will be required to identify the core functionality of your proposed design and how it meets the interaction requirements.

So bring your plans and plots and schemes; we've got the mechanics if you've got the dreams!

Professor Elemental (FIRST Steamworks-Steampowered, 2017)

First Checkpoint:

On or before **11/7/25**, you must submit a schematic of at least the core functionality initially identified on **11/4/25** and a refined set of events with details on the responses. Modifications to the core functionality may take place up to this point. A KiCad schematic (printed to pdf or added to a word document) describing your core functionality should be uploaded to Gradescope. Only one team member needs to submit your checkpoint documentation.

Second Checkpoint:

On or before **11/14/25** you will be required to demonstrate a minimal level of function:

1. The hardware & software necessary to do each of the following:
 - (a) sense inputs (at least 3 user inputs)
 - (b) make decisions (state machine with at least 3 states driven by keyboard input)
 - (c) implement electromechanical actuation and user feedback
2. Submission of a KiCad schematic of your circuit will also be required.

Third Checkpoint:

On **11/19/25** you will be required to demonstrate integrated functionality of all sensing inputs, plus software and timing, plus activating all actuators that will be required. In other words, everything should be complete with the exception of improvements in user experience and fit, finish, and appearance.

Grading Session:

On **11/20/25** from **1:00 pm** to **5:00 pm** you will be required to demonstrate your fully integrated and finished TEM.

Public Presentation:

This will take place on **11/20/24** starting at **7:00 pm** in the Atrium of Building 550. At this event, members of the public will be encouraged to act as confused Karl'nival goers, and will interact with your TEMs.

Report:

Draft due on **12/1/25** by **4:00 pm**. The final version (with revisions incorporated) is due by **5:00 pm** on **12/5/25**.

Evaluation

Performance Testing Procedures:

All TEMs will be tested by a demonstration, performed by a team member, that should show all of the possible user interactions.

Grading Session Presentation:

Each team should prepare a 30 second (no more) presentation to introduce their TEM. This presentation should highlight the unique features of the design, not the circuit or software details. As an example, think back to the xylophone descriptions that were played on the first day of class. You will be setting up your TEM, one at a time, and delivering your presentation to the teaching team in the SPDL between 1:00 pm & 5:00 pm on the day of the presentations. During this time each team and their TEM will be photographed. Starting at 5:00 pm you will move your TEM into the 550 Atrium for the public presentation, which will begin at 7:00 pm.

Grading Criteria:

- **Concept (20 %)** This will be based on the technical merit of the design and coding of your project. Included in this grade will be evaluation of the appropriateness of the solution, as well as innovative hardware, software and use of physical principles in the solution.
- **Implementation (20 %)** This will be based on the prototype displayed at the evaluation session. Included in this grade will be evaluation of the physical appearance of the prototype and quality of construction. We will concentrate heavily on craftsmanship and finished appearance.
- **Performance (40 %)** Half of this (20 %) will be based on the results of the checkpoints, the other half will be based on the results of the performance testing during the evaluation session. Full performance credit will be given only if the TEM works on the first attempt during the grading session. Performance will be judged first on the ability to demonstrate the core functionality and second on any embellishments to the core functionality. **To earn the performance points, you must demonstrate at least the core functionality.**
- **Report (10 %)** Preliminary project reports are due **12/1/25 at 4:00 pm**. The report should be in the form of a stand-alone web site and must include schematics, pseudo-code, header & code listings, dimensioned sketches/drawings showing relative scale, a complete Bill-of-Materials (BOM) for the project as well as a 1 page description of function and a “Gems of Wisdom for future generations of 218ers” page. The actual website must be hosted and you must submit the URL to your site in the specified spreadsheet. It is critical that the URL to your report be in the spreadsheet on time so that the peer reviewing team will have an adequate opportunity to review it before class the following day. Final versions of the reports, incorporating the review comments are due by **5:00 pm on 12/5/25**.
- **Report Review (10 %)** These points will be awarded based on the thoroughness of your review of your partner team’s report. Read the explanations, do they make sense? Review the circuits, do they look like they should work? Could this project realistically be built for \$200? If, during grading, we find things that don’t make sense or circuits that won’t work we will consult your review. If the review caught them, then the team will lose points on their report. If the reviewers missed it, then they will lose points for their review. The report review should be submitted on Gradescope by **5:00 pm on 12/5/25**.

Resources

Websites:

[SparkFun \(www.sparkfun.com\)](http://www.sparkfun.com)
[Jameco \(www.jameco.com\)](http://www.jameco.com)
[Newark \(www.newark.com\)](http://www.newark.com)
[Adafruit \(www.adafruit.com\)](http://www.adafruit.com)
[Digi-Key \(www.digikay.com\)](http://www.digikay.com)
[HobbyKing \(www.hobbyking.com\)](http://www.hobbyking.com)

[Seed Studio \(www.seeedstudio.com\)](http://www.seeedstudio.com)
[Mouser \(www.mouser.com\)](http://www.mouser.com)
[Ponoko \(www.ponoko.com\)](http://www.ponoko.com)
[Hackaday \(www.hackaday.com\)](http://www.hackaday.com)
[McMaster-Carr \(www.mcmaster.com\)](http://www.mcmaster.com)
[ServoCity \(www.servocity.com\)](http://www.servocity.com)

You may also find [PlantUML](#) and [PlantText](#) helpful for creating diagrams of various types.

Local Stores:

[Anchor Electronics](#) in Santa Clara
[Sheldon's Hobbies](#) in San Jose
[Jameco](#) in Belmont
[TAP Plastics](#) in San Mateo

Gems of Wisdom:

Be sure to check out [The ME218 Archive](#) for guidance from past generations.

Revision History

Revision 0: Initial version. (10/30/25)